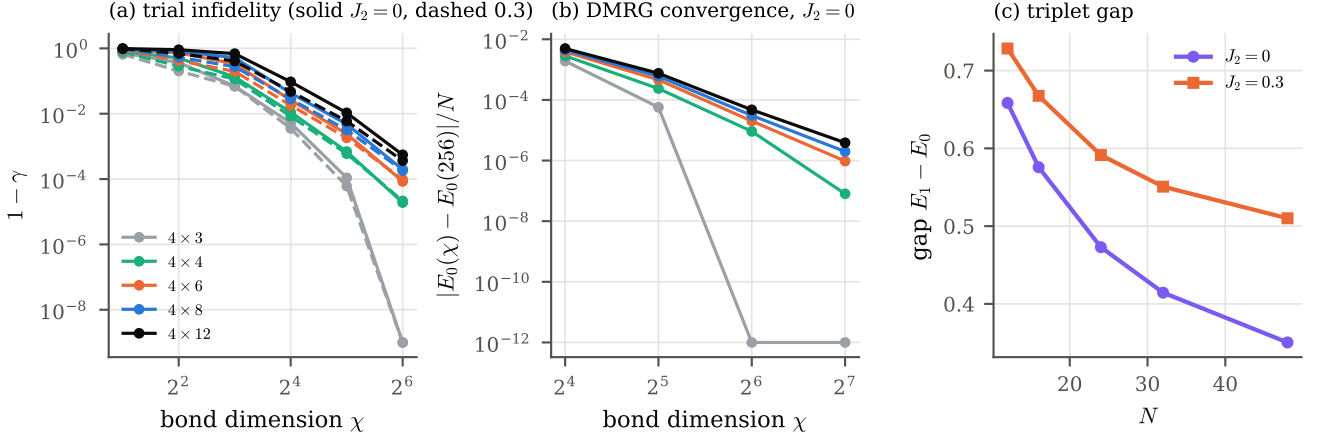
(a) trial infidelity (solid $J_2 = 0$, dashed 0.3)
$1 - \gamma$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
4×3
4×4
4×6
4×8
4×12
2^{2}
2^{4}
2^{6}
bond dimension χ
(b) DMRG convergence, $J_2 = 0$
$|E_0(\chi) - E_0(256)|/N$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
2^{4}
2^{5}
2^{6}
2^{7}
bond dimension χ
(c) triplet gap
gap $E_1 - E_0$
0.7
0.6
0.5
0.4
$J_2 = 0$
$J_2 = 0.3$
20
30
40
N